

Byeong-Il Ham

✉ byeongil_myLastName@kaist.ac.kr 📍 Daejeon, Republic of Korea

Education

Korea Advanced Institute of Science and Technology , Robotics	Daejeon, KR 2024 – present
Korea Advanced Institute of Science and Technology , Robotics Robust Swing Leg Planning Algorithm for Quadruped Robots Using Dual Convex Model-Predictive Control	Daejeon, KR 2022 – 2024
University of Kwangwoon , School of Robotics	Seoul, KR 2016 – 2022

Experience

IEEE , Graduate Student Member	Mar 2025 – present 1 year 6 months
IEEE Robotics and Automation Society , Technical Committee Member - Model-Based Optimization for Robotics - Humanoid Robotics - Robot Control - Robot Learning	Jan 2026 – present 8 months

Awards

Best System Paper Award Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, Jeonguk Kang, Minseong Choi, Keun-Ha Choi, Kyung-Soo Kim	Dec 2024
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Publications

Accelerating Human-Aware Robot Trajectory Generation via Diffusion and Consistency Distillation

Byeong-Il Ham, Hyun-Bin Kim, Kyung-Soo Kim

Dual-MPC Footstep Planning for Robust Quadruped Locomotion

Byeong-Il Ham, Hyun-Bin Kim, Jeonguk Kang, Keun Ha Choi, Kyung-Soo Kim

doi.org/10.1109/TMECH.2026.3720700

Ground-Aware Octree-A* Hybrid Path Planning for Memory-Efficient 3D Navigation of Ground Vehicles

Byeong-Il Ham, Hyun-Bin Kim, Kyung-Soo Kim

ieeexplore.ieee.org/abstract/document/11301310

Adaptive Gait Pattern Switching under External Disturbances Using Multi-Modal MPC

Jeonguk Kang, Byeong-Il Ham, Seungho Han, Hyun-Bin Kim, Kyung-Soo Kim

ieeexplore.ieee.org/document/11217651

Compact and Optimized Optical Six-Axis Force/Torque Sensor for Legged Robots

Hyun-Bin Kim, Byeong-Il Ham, Keun-Ha Choi, Kyung-Soo Kim

ieeexplore.ieee.org/document/11194133

A Miniature High-Resolution Tension Sensor Based on a Photo-Reflector for Robotic Hands and Grippers

Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim

ieeexplore.ieee.org/abstract/document/11150551

Development of Remote Piping Inspection System with Dual-Mode Locomotion Quadruped Robot

Hyun-Bin Kim, Chanseok Kim, Byeong-Il Ham, Jeonguk Kang, Minseong Choi, Keun-Ha Choi, Kyung-Soo Kim

link.springer.com/chapter/10.1007/978-3-031-92011-0_25

Temperature Compensation Method for a Six-Axis Force/Torque Sensor Using a Gated Recurrent Unit

Hyun-Bin Kim, Seokju Lee, Byeong-Il Ham, Keun-Ha Choi, Kyung-Soo Kim

ieeexplore.ieee.org/document/11021311

External force adaptive control in legged robots through footstep optimization and disturbance feedback

Jeonguk Kang, Hyun-Bin Kim, Byeong-Il Ham, Kyung-Soo Kim

ieeexplore.ieee.org/document/10731682

Reviewing

Springer International Journal of Control, Automation, and Systems (IJCAS), Reviewer 2026

International Conference on Control, Automation, and Systems (ICCAS), Reviewer 2026

Projects

Development of an AI Robot Platform and Standard System for Seawater Pipeline Inspection Apr 2022 – Dec 2022

Developed robot platform technologies and standard systems for seawater pipeline inspection.

- Korea Hydro Nuclear Power
- Embedded Systems Engineering

Development of high-speed precision driving algorithm for intelligent active protection technology June 2023 – Dec 2023

Development of active protection technology system.

- Hanwha Aerospace
- Project Manager

DRB-KAIST SketchTheFuture Research Center July 2025 – present

Robotic immersive interaction platform for future mobility space design.

- DRB-Industrial
- Project Manager since 2025

Single-Person Multi-Purpose Autonomous Micro-Mobility for Factory and Campus Feb 2024 – present

Developed autonomous micro-mobility platforms and embedded control systems for factory and campus environments.

- DRB-Industrial
- Project Manager

Skills

Tools

Interests

Robotics

Languages

Korean

Native speaker

English

Intermediate